

What Do Chemical Language Models Learn? Interpreting Molecular Transformers with Sparse Autoencoders

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Abstract

Molecular transformers trained on millions of molecules predict chemistry accurately, yet how they represent it internally remains opaque. Sparse autoencoders can disentangle a model’s overlapping internal signals into readable features, but interpreting these features traditionally relies on observational association—identifying structures common to a feature’s activating molecules. Here, we propose InterpretMolT, a method that pairs sparse autoencoders with counterfactual editing to causally test these structural associations. Applying this framework across six molecular models, we find that while 86–100% of features associate with specific structures, 38–46% of these associations fail causal necessity tests. Crucially, our testing reveals a stark functional dichotomy: a sparse minority of ”dense” features performs over 93% of the representational reconstruction but exhibits the weakest causal dependence on their associated structures. Conversely, the vast majority of ”specific” features genuinely read chemical structures—predominantly from peripheral atoms rather than the central scaffold—but contribute minimally to the model’s overall workload. By separating mere correlation from true causal dependence, InterpretMolT demonstrates that a model’s most functionally consequential features are often the least structurally interpretable, establishing a rigorous new foundation for auditing chemical language models.

Keywords: molecular transformers, sparse autoencoders, cheminformatics, mechanistic interpretability, structural association, causal testing, explainability

1. Introduction

A molecule can be written as a string using the SMILES notation, which spells out its atoms and the bonds between them [1]. In recent years the transformer, the neural network behind large language models, has been trained to read molecules written this way, learning from tens of millions of examples collected in public chemical databases such as ZINC and

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PubChem [2, 3, 4]. These molecular language models now predict properties such as toxicity and solubility, among many others, with high accuracy, and they are increasingly used to speed up the earliest stages of drug discovery, where millions of candidate compounds must be sifted quickly [5, 6, 7]. When one of these models flags a molecule as toxic, though, we cannot see what about its structure drove that verdict, nor whether the model has grasped the underlying chemistry or has simply latched onto patterns in its training data that happen to track the right answers. This gap carries practical weight. It leaves researchers choosing between models on their benchmark scores alone, with no clear view of what each has actually learned, and it sits uneasily beside a growing expectation, voiced in regulatory guidance from the U.S. Food and Drug Administration and the European Medicines Agency on artificial intelligence in drug development, that anyone deploying such a model should be able to explain how it reaches its decisions [8, 9].

One response has been to borrow the tools already built for understanding models in image and text tasks, and to turn them on chemistry. These tools work one molecule at a time: given a single prediction, they point to the parts of the input that mattered most, marking which atoms pushed the model toward its answer. Some read off the internal weights the model places on each part of the molecule as it processes the text [7, 10]; others trace a prediction back to individual atoms by measuring how much each one shifts the output [11, 12]. Two things limit how far this gets us. The weights a model places on its input have been shown to be an unreliable guide to which parts truly drove a decision [13]. And an explanation built for one molecule says nothing about the internal parts the model reuses across every molecule it sees, the recurring building blocks of what it has learned. Reaching those parts calls for a different kind of access.

Why are those reusable internal parts so hard to reach in the first place? As a model learns, it stores far more distinct ideas than it has internal signals to hold them, so it packs several unrelated ideas into each signal and lets them share [14]. A single signal, read on its own, then answers to a jumble of unrelated inputs at once, and no one signal stands cleanly for any one thing [15]. A sparse autoencoder is a second, smaller network trained to undo this packing [16, 17]. It learns to re-express the model’s tangled internal signals as a much larger set of features, and to keep only a handful of them active for any one molecule [18], so that each feature is ideally freed to respond to a single recognisable pattern. What the model had folded together is pulled back apart into separate, inspectable parts, each one a candidate piece of what the model has learned to notice.

Having these features in hand still leaves the question of what each one responds to. The natural way to answer it is to collect the molecules that switch a feature on most strongly and look for the piece of chemistry they share; when nearly all of them carry the same ring or chemical group, that shared structure becomes the feature’s likely meaning. Such a reading rests on structure and feature merely appearing together, and a structure that appears alongside a feature need not be one the feature depends on: the feature might switch on whenever a certain ring is present and still not need it, the ring simply travelling with some other detail that is the real trigger. To tell the two apart, there is one direct move, take the structure out of a molecule that switches the feature on, and watch whether the feature falls quiet or fires just the same. Here chemistry adds an obstacle, since a molecule written as a line of text tends to break when altered: change it carelessly and the line no longer spells out any real molecule [1], so the test must remove the target structure while

leaving a whole molecule behind.

Sparse autoencoders have already been used to read the features of molecular models this same way, though so far only from inside the model. One study switched a feature off and saw the structure change in the molecule the model then produced [19]; another switched off parts of the model to find which ones keep a molecule’s written form valid [20]. Each looked at a single model, and neither took the structure a feature responds to out of a molecule to see whether the feature then falls quiet. Nor is this question particular to chemistry: in models that read images, a feature’s meaning is read the same way, from the parts of an image that switch it on most strongly, and there a part can switch a feature on while merely travelling with it, something else being the real trigger [21].

Here we introduce **InterpretMolT**, a method that moves beyond observational correlation by reading each separated feature from the molecules that activate it and then testing that reading directly. We train a sparse autoencoder on the signals inside a model to obtain the features, then gather each feature’s strongest activating molecules and look for a structure they share. That shared structure can be a scaffold, the central ring framework a molecule is built around [22], or a functional group, a small cluster of atoms that gives a molecule a particular chemical behaviour, such as an alcohol or an acid [23]. A structure that appears among these molecules far more often than across molecules in general is a candidate for what the feature reads, and we carry forward the strongest one to test. The test makes the smallest change a molecule allows: we swap a single atom for another of similar size, so the molecule stays readable and the target structure is the only thing removed. Sending the edited molecule back through the model shows what happens: a feature that relied on the structure goes quiet, while one that merely sat alongside it keeps firing. Since altering any atom disturbs a molecule slightly, we pair each edit with a control change elsewhere in the same molecule that leaves the target structure intact, so a larger drop from the edit that removes the structure than from the control marks it as one the feature actually reads. The same paired edit also shows where a feature reads, since placing one change on the scaffold and the other on the surrounding atoms reveals whether it draws on the central framework or the outer parts. We apply **InterpretMolT** to six pretrained molecular models spanning three different architectures, drawing the molecules from the MOSES collection of drug-like compounds [24].

Across all six models, almost every feature lined up with a structure: in each model, between 86% and 100% carried an association strong enough to stand out from chance. When we put those associations to the direct test, however, many gave way. In 38% to 46% of the cases we could test, removing the structure left the feature firing much as before, so it had been leaning on something else all along. Furthermore, these causal tests exposed a stark dichotomy in how these models distribute their efforts. A tiny minority of “dense” features handles over 93% of the representation-rebuilding workload, yet these are the exact features that show the weakest causal reliance on specific chemical groups. Meanwhile, the vast majority of “specific” features—which definitively track structures—do only a fraction of the work. The same edits also showed where on a molecule each feature reads, and here the models agreed: among features that clearly favoured one region, between 78% and 95% drew on a molecule’s outer atoms far more than on its central framework. To our knowledge, this is the first study to test whether the structures a molecular language model’s features associate with are ones those features actually need in order to activate, and the first to prove

that a model’s most load-bearing features are paradoxically its least structurally dependent. Pairing each association with a direct test of whether the feature depends on it gives a firmer footing for judging what these models have learned, and for trusting a feature’s reading when a decision turns on it. The sections that follow trace this from the shape of the feature space, through the structures each feature is tied to, to the tests that separate the ones a feature reads from the ones it merely appears beside, and show where on a molecule it reads.

2. Methods

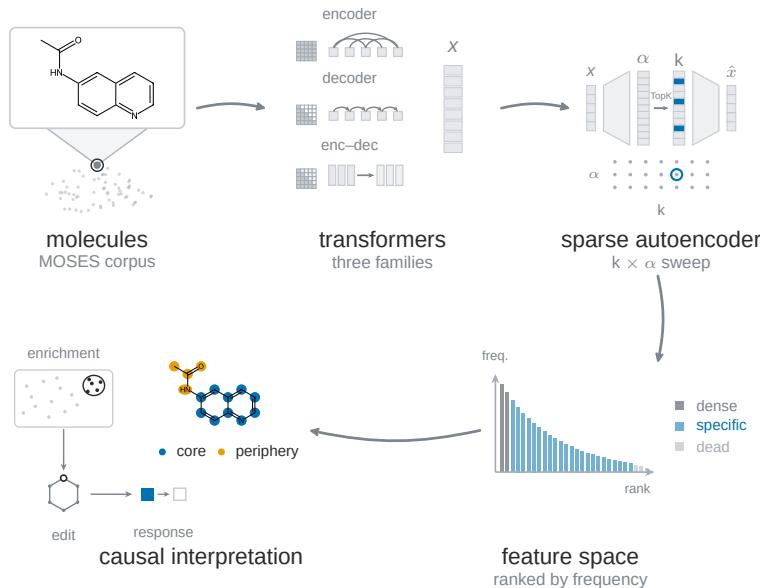


Figure 1: **Analysis pipeline.** Interpreting molecular-transformer features through sparse autoencoders, from embedding extraction to causal interpretation.

Our method proceeds in four stages, each a precondition for the next: we extract molecular embeddings from each pretrained transformer, train a sparse autoencoder (SAE) on them and select its hyperparameters per model, characterise the resulting feature dictionary, and interpret individual features through structural association and causal testing.

2.1. Models, Data and Embeddings

2.1.1. Model Selection

We analysed six pretrained molecular transformers drawn from three architecture families: encoder-only (MoLFormer [5], ChemBERTa-MLM [6]), which encode SMILES bidirectionally; decoder-only (GP-MoLFormer [25], MolGPT [26]), which encode them autoregressively; and encoder-decoder (BARTSmiles [27], ZINC-t5-v2 [28]), which combine both. All six are publicly available and were pretrained on large public molecular corpora, predominantly ZINC [3] and PubChem [4]. Analysing this span, rather than a single model, lets us analyse shared findings with architectures rather than to one model’s idiosyncrasies; throughout, we treat architecture family as a grouping label for reporting and comparison, not as

Table 1: **Pretrained molecular transformer models used.**

Model	Architecture	Layers	Dimensions	Parameters	Pretraining data
MolFormer	Encoder	12	768	44.3M	ZINC + PubChem (~ 100 M)
ChemBERTa-MLM	Encoder	3	384	3.4M	PubChem (77M)
GP-MolFormer	Decoder	12	768	46.7M	ZINC + PubChem (unique)
MolGPT	Decoder	8	256	6.9M	ZINC15
BARTSmiles	Encoder-decoder	6+6	1024	354M	ZINC20 (1.7B)
ZINC-t5-v2	Encoder-decoder	12+12	768	198M	ZINC

an asserted causal variable. Each model’s depth, representation dimensionality, parameter count and pretraining corpus are given in Table 1.

2.1.2. Dataset and Split Policy

We applied each trained SAE to molecular embeddings from MOSES [24], a benchmark of approximately 1.9 million drug-like molecules with three standard train, test and scaffold-test splits; the scaffold-test split (`test_scaffolds`) is constructed so that its Bemis–Murcko scaffolds do not appear in the train split [22]. The three splits serve distinct roles. SAEs were trained on the `train` split (1,584,664 molecules). The hyperparameter sweep was evaluated, and the feature space characterised, on the in-distribution `test` split (176,075 molecules), so that reconstruction fidelity and sparsity coincide with the quantities used for hyperparameter selection, which are computed on the same split. All feature interpretation, building the structural vocabulary, testing for association, and testing for necessity, was done on `test_scaffolds` (176,226 molecules). Characterisation and interpretation therefore run on different splits. What carries a feature from one to the other is only how it fires, how often, how much it contributes, and how dense it is, never anything about scaffold identity. Nothing about the held-out scaffolds, then, shapes either the SAE settings or which features go forward to interpretation. MOSES is distributed pre-filtered for chemical validity and drug-likeness; we applied no further canonicalisation and removed only molecules that failed to parse.

2.1.3. Embedding Extraction

To get one fixed-size embedding per molecule, we passed each split’s SMILES through its pretrained transformer and took the hidden state at the model’s final layer, one vector per token. We averaged these into a single vector per molecule, using the attention mask to leave out the padding added to fill out shorter sequences. For the two encoder–decoder models, this final layer was the encoder’s, not the decoder’s. Each model’s hidden state has its own width, so each model’s embedding has its own fixed size.

2.2. Sparse Autoencoders

2.2.1. Architecture

For each of the six models we trained one TopK sparse autoencoder on its pooled embeddings, following Gao et al. [18]. Each autoencoder takes a d_{model} -dimensional embedding and expands it into a much larger, overcomplete code of $d_{\text{hidden}} = \alpha d_{\text{model}}$ latents, one candidate feature per number, of which only k are ever switched on for a given molecule (Fig. 2). Both

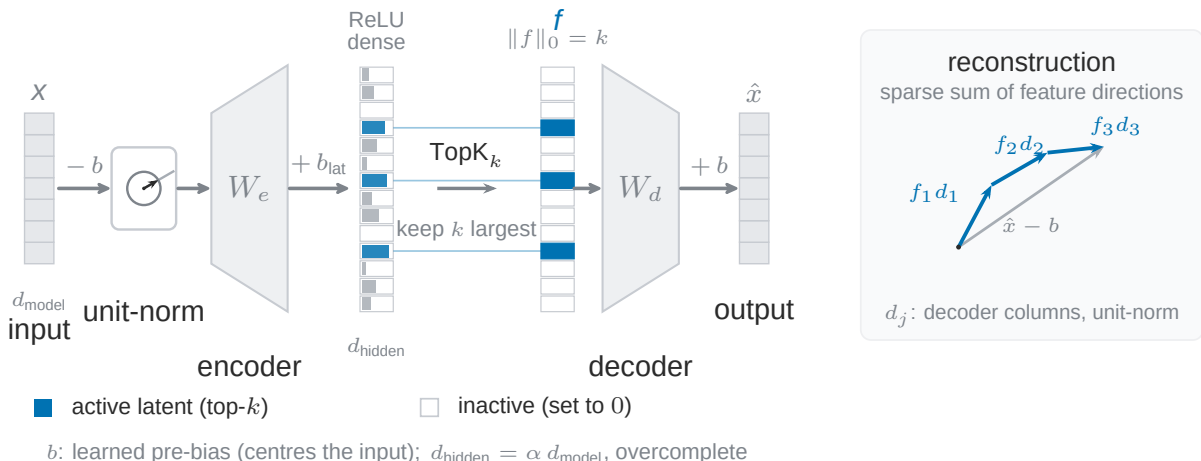


Figure 2: **TopK Sparse Autoencoder Architecture and Forward Pass.** The input embedding is encoded to an overcomplete latent code, of which TopK_k keeps the k largest activations before the decoder reconstructs the input.

α and k were set separately for each model (Hyperparameter Sweep and Selection, below). The autoencoder has four learnable parts: a pre-bias, an encoder weight, a latent bias, and a decoder weight; neither the encoder nor the decoder carries a bias of its own. Encoding works in three steps: the embedding is centred by subtracting the pre-bias and rescaled to unit length, then projected through the encoder weight and latent bias, then passed through a ReLU, which zeros out anything negative. What survives is cut down further to only its k largest entries; everything else is set to zero. Decoding reverses only this last step: the sparse code is passed through the decoder weight, and the pre-bias is added back:

$$\mathbf{f} = \text{TopK}_k \left(\text{ReLU} \left(\mathbf{W}_e \frac{\mathbf{x} - \mathbf{b}}{\|\mathbf{x} - \mathbf{b}\|_2} + \mathbf{b}_{\text{lat}} \right) \right), \quad \hat{\mathbf{x}} = \mathbf{W}_d \mathbf{f} + \mathbf{b}, \quad (1)$$

where TopK_k keeps the k largest values and zeros the rest, so exactly k features are switched on for every molecule, by construction. The decoder was set up first, and the encoder was initialised as its transpose, so the two began as mirror images of each other. After every training step, the decoder’s columns, each one the direction in embedding space that a single feature points toward, were rescaled back to unit length. This keeps what tells one feature apart from another to be its direction, not how large its weights happen to grow.

2.2.2. Training

Each autoencoder was trained on its model’s **train** embeddings, for 100,000 steps, in batches of 128, using Adam [29] with a learning rate of 10^{-4} . The learning rate ramped up over the first 1000 steps, then decayed back down to zero. Before training began, the pre-bias was set to the average embedding across the training set, so the autoencoder started from a sensible centre rather than from zero.

Training pushed toward two goals at once. The main goal was reconstruction: make $\hat{\mathbf{x}}$ close to $\mathbf{x}$. But since only k features switch on at a time, some features can go long stretches without ever being chosen, stop receiving useful gradient, and effectively die. To stop this, a second term gave dead features a job of their own: whatever the active features failed to

Table 2: **Selected per-model SAE configuration and dictionary health.** Settings were chosen by the deterministic selection pass on the `test` split.

Model	Family	d_{model}	k^*	α^*	d_{hidden}^*	k^*/d_{hidden}^*	FVE	alive
MoLFormer	Encoder-only	768	512	8	6144	0.083	0.985	0.966
ChemBERTa-MLM	Encoder-only	384	256	4	1536	0.167	0.993	0.956
GP-MoLFormer	Decoder-only	768	1024	4	3072	0.333	0.995	0.964
MolGPT	Decoder-only	256	256	4	1024	0.250	0.992	0.995
BARTSmiles	Encoder-decoder	1024	768	4	4096	0.188	0.993	0.977
ZINC-t5-v2	Encoder-decoder	768	512	8	6144	0.083	0.988	0.960

reconstruct, the residual $\mathbf{e} = \mathbf{x} - \hat{\mathbf{x}}$, was handed to the features that had gone quiet, and they were asked to reconstruct that leftover error instead:

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda_{\text{aux}} \|\mathbf{e} - \hat{\mathbf{e}}\|_2^2, \quad \mathbf{e} = \mathbf{x} - \hat{\mathbf{x}}, \quad (2)$$

where $\hat{\mathbf{e}}$ is the residual reconstructed from only the top-`auxk` features that had stayed off over a recent stretch of training. We set $\lambda_{\text{aux}} = 1$ and `auxk` = 512 for every model.

2.2.3. Hyperparameter Sweep and Selection

Two settings shape each autoencoder: the sparsity k , how many features switch on, and the expansion factor α , which sets the dictionary size, $d_{\text{hidden}} = \alpha d_{\text{model}}$. Rather than pick these by hand, we swept a grid and let the data decide. For each model, we trained one autoencoder at every combination of twelve sparsity levels ($k = 16, 32, 48, 64, 128, 192, 256, 384, 512, 768, 1024, 1280$) and three expansion factors ($\alpha = 4, 8, 16$), giving 216 combinations across the six models, using the training setup above, and evaluated each on the in-distribution `test` split.

Three numbers were measured per combination. Fraction of variance explained (FVE) is how faithfully the autoencoder reconstructs its input:

$$\text{FVE} = 1 - \frac{\sum_j \text{Var}(\hat{x}_j - x_j)}{\sum_j \text{Var}(x_j)}, \quad (3)$$

Higher means less is lost. The alive fraction is the share of features that ever switch on for at least one test molecule; features that never do are dead weight. The duplicate fraction is the share of features pointing in nearly the same direction as another, with higher values meaning two features doing the same job instead of two different ones.

Selection followed one fixed rule, applied separately to each model. Combinations with $k \geq d_{\text{hidden}}$ were dropped first, since with that few features a k -sparse code isn’t possible. Among what remained, we kept only combinations reaching at least 99% of that model’s own best FVE, then dropped any with an alive fraction below 0.95 or a duplicate fraction above 0.05. Among the survivors, we picked the smallest k , since sparsity is what makes a feature interpretable in the first place, breaking ties by the smallest α . Because the fidelity bar is set relative to each model’s own best score, it adapts to each model rather than imposing one number on all six; the other two thresholds are the only fixed choices, and their effect is checked directly below. We then asked whether one (k, α) could work for all six models

at once. None did, so no shared setting exists, and each model keeps its own. The rule is shown per model in Supplementary Fig. 1, and each model’s chosen k , α and dictionary size are listed in Table 2.

A separate control sweep, fixing the same absolute dictionary size across all six models instead of the same expansion factor, confirmed that fidelity tracks k and not which convention is used, while forcing one shared dictionary size makes the narrower models worse off, degrading their alive fraction (Supplementary Fig. 2a). Selection was therefore based on the expansion sweep. Sensitivity of this divergence result to the two threshold values is reported in Supplementary Fig. 2b.

2.3. Feature-Space Characterisation

Before asking what any single feature means, we characterised each trained dictionary as a whole, on the test split; this section describes the shape of the dictionary, not yet what it captures chemically. Because this is the same split the SAEs were selected on, the fidelity numbers we would recompute here are unchanged from the sweep, so we do not repeat them.

We first worked out which features have anything to say. A feature that never switches on anywhere in the test split is dead; it has no molecules to describe and nothing to contribute. Every feature that does switch on is alive, and only alive features are studied from here on. Among the alive features, we separated dense from specific: a feature switching on for at least half of all test molecules is dense, and one switching on less often is specific. This half-way line is a reporting choice, not a hard boundary in the data, since every alive feature is analysed regardless of which side it falls on, so we also checked how the dense count moves if the line is drawn anywhere from 0.3 to 0.9 (Supplementary Fig. 3).

We then asked how much rebuilding work each feature does. A feature’s contribution is how often it switches on, summed over all molecules, times the length of the direction it adds back when decoding. This is not a variance decomposition, because the dictionary has far more features than the embedding has dimensions, so their directions overlap and their individual contributions cannot be added up cleanly, but it is enough to rank features by how much weight each carries. We summarised this ranking with a Gini coefficient (how unevenly the weight is shared) and by counting how many top features, added in order by contribution, it takes to reach half the dictionary’s total contribution.

We next checked whether the dictionary really is a set of distinct features, rather than many near-copies of a few. For every alive feature’s decoder direction, we found its closest neighbour among the others by cosine similarity, and counted how many pairs were near-identical (similarity above thresholds from 0.7 to 0.95). We also measured how many directions the alive features effectively spread across, compared to the same count of random directions. Because the dictionary has far more features than the embedding has dimensions, some closeness between directions is unavoidable regardless of training, so every one of these numbers was compared against the same measurement run on the same number of random directions, and is read only relative to that baseline.

We ranked every alive feature by how often it switches on, and checked how closely this ranking agreed with the contribution ranking (Spearman correlation). The dense-or-specific label from earlier is carried forward on every feature as a note, not used to filter anything out.

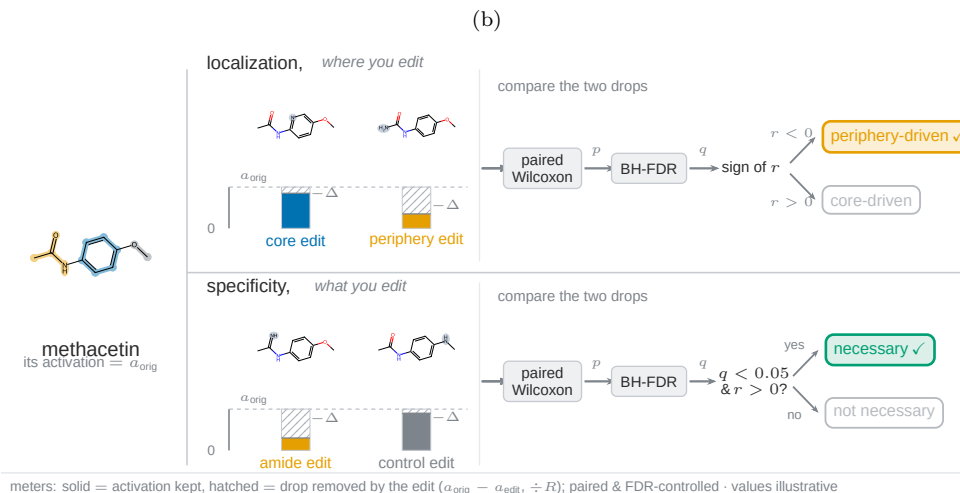
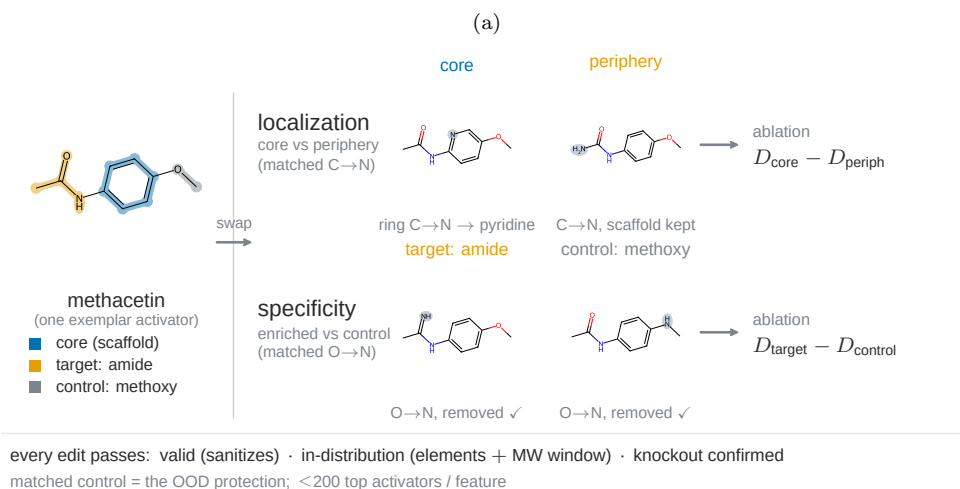
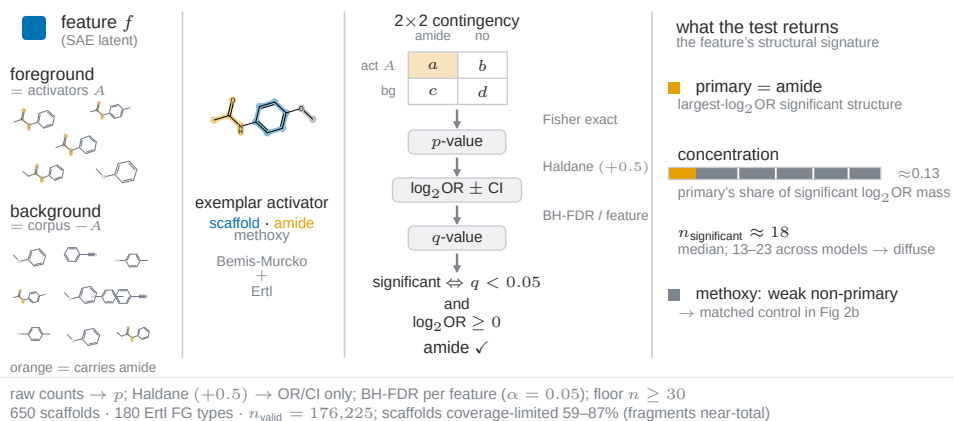


Figure 3: From association to necessity, on one exemplar feature. (a) A 2×2 Fisher exact test (Haldane–Anscombe $\log_2$ odds ratio, per-feature BH-FDR) flags the feature’s primary signature. (b) Two matched single-atom edits follow: core-vs-periphery (locus) and target-vs-control (necessity). (c) Paired activation drops are compared by Wilcoxon signed-rank test [30] (rank-biserial r , BH-FDR): its sign sets the locus, and a significant positive r marks the structure necessary.

2.4. Feature Interpretation

2.4.1. Structural Vocabulary

Interpretation tests each feature against a fixed structural vocabulary, which we built once on the `test_scaffolds` corpus before any feature was examined. The vocabulary is purely structural, a function of the SMILES and RDKit [31] alone with no SAE or model involved, and is therefore shared unchanged across all six models. It has two granularities. At scaffold granularity, each molecule contributes its Bemis–Murcko scaffold [22]. At functional-group granularity, each molecule contributes the set of functional-group types identified by the Ertl algorithm [23] (vended verbatim from RDKit’s contributed implementation). Stereochemistry was removed before deriving either key, so the vocabulary entries are chemical classes rather than stereo-split variants; the molecules the models embedded were never altered.

A structure entered the vocabulary if it occurred in at least 30 valid molecules, a floor read from the rank-frequency distribution of each granularity rather than asserted. Over the 176,225 molecules, this retained 650 scaffolds and 180 functional-group types. Scaffold coverage is intrinsically limited on a scaffold-split corpus, since most molecules carry a near-unique scaffold that no floor can retain, so the fraction of molecules whose scaffold is in-vocabulary is reported alongside every scaffold result; functional-group coverage is near-complete and is not gated (Supplementary Fig. 4).

2.4.2. Structural Association

Association asks, for one feature at a time, which structures turn up among the molecules that switch it on more often than they do everywhere else. What it finds are candidates only, not yet the structures a feature actually needs; that is what the causal leg decides next (Fig. 3). For a feature and a structure from the vocabulary, we split the valid molecules two ways at once, by whether each one switches the feature on and by whether it carries the structure. This gives four counts, read directly as exact integers from the per-molecule vocabulary tables, never estimated (Fig. 3a). We then asked whether the structure showed up among the feature’s activators more than chance allows, using a one-sided Fisher exact test, computed as a hypergeometric survival function on these four counts. To size an association and not just flag it, we report an odds ratio under a Haldane–Anscombe correction [32, 33], which adds half a count to each of the four cells so the ratio stays finite even when one of them is zero, alongside its Woolf 95% confidence interval [34]; this correction shapes only the odds ratio and its interval, never the test that decides significance, which runs on the raw counts. Scaffolds and functional groups were tested the same way throughout, with nothing in the procedure treating one differently from the other.

A feature is tested against every structure in the vocabulary at once, so its many results cannot be read alone; each one raises the chance of a false hit by luck. We treated a feature’s full set of tests as one family and corrected it with the Benjamini–Hochberg procedure [35]: a structure counts as significant when its corrected q -value falls below 0.05 and its odds ratio points the enriching way, at or above one. Among a feature’s significant structures, the one with the largest odds ratio is its primary signature, and we record what share of the feature’s total association this one structure accounts for, its concentration: close to one means a single structure dominates what the feature is tied to, close to zero means the feature’s associations are spread thinly across many. Scaffolds carry one further complication. Because the held-out molecules mostly carry rare or one-off scaffolds, most cannot be tested

at all, so every scaffold result comes with the share of the feature’s activators that even carry a scaffold common enough to test against.

Four numbers, computed the same way for both granularities, carry these associations into the results that follow. Signature coverage is the share of features that land on at least one significant structure. The number of significant structures per feature is reported as a full distribution, not a single average, since this varies widely from one feature to the next. The primary signature’s odds ratio and concentration are as defined above. And primary-structure capture turns the question around: of every molecule in the corpus that carries a feature’s primary structure, what share of them does the feature switch on?

2.4.3. *Counterfactual Edits*

A structure cannot simply be deleted from a molecule; what would be left behind is not a molecule at all. Every edit we make instead changes one atom into another, keeping every bond in place, so the size and shape of the molecule stay exactly as they were and only the identity of a single atom moves. This is the same single-atom edit used throughout the paper, applied here to build the paired molecules the causal tests compare (Fig. 3b). Because the two edits in a pair differ only in which atom is changed and never in how many atoms are changed, a difference in how the feature responds cannot be attributed to the size of the edit; it can only reflect what was changed and where.

Every swap moves an atom among carbon, nitrogen, oxygen and sulfur, the four elements common enough in these molecules that exchanging one for another leaves a valid molecule behind. Where a molecule allows more than one such swap, we chose among them in a fixed order, so the same molecule is always edited the same way.

Two paired contrasts are built from this one swap. Necessity, put to features carrying a functional-group signature, targets that whole group: every atom capable of removing an instance of it is swapped, one swap per instance, until every occurrence of the group is confirmed gone from the molecule. Its partner is the same operation carried out on a different functional-group type, present in the same molecule, chosen to disturb about as many atoms as the target did but left off the feature’s list of associated structures; its only job is to leave the target group standing while unsettling the molecule by a comparable amount elsewhere, to serve as a control. Localisation, put to every alive feature, swaps one atom on the Bemis–Murcko scaffold against one atom off it, so one edit changes the scaffold and the other leaves it standing; wherever a swap of the same kind is available in both places, we use that same swap in both, so the two edits differ only in where they fall, not in what they do.

Not every attempted swap succeeds. An edit is kept only if the resulting molecule is chemically valid and stays within the range of elements and molecular weights already seen in the corpus. For each edit we report a validity rate, the share of attempted swaps that met both conditions, as a check on how often the editing procedure could actually be carried out (Supplementary Fig. 5).

2.4.4. *Causal Tests: Necessity and Locus*

Both questions are put to the same kind of comparison, though each draws on different molecules and reads the result differently. For every molecule that switches a feature on, we passed the molecule’s two edited versions back through the model alongside the unedited

original, and measured how far each edit dropped the feature’s activation relative to that original, its drop. What the test then asks is not whether either drop alone is significant, but whether one of the two drops is consistently the larger across the molecules gathered for that feature (Fig. 3c).

Necessity, put only to features carrying a functional-group signature, compares the drop from knocking out the feature’s primary group against the drop from knocking out a control group instead. A functional group is defined by a handful of atoms sitting in one place, so swapping just those atoms clears the group while every other atom and bond stays exactly as it was. A scaffold offers no such handful to swap: it is the ring framework the rest of the molecule is attached to, so there is no small set of atoms whose replacement would take the scaffold away and still leave the rest of the molecule standing on it. A molecule enters the necessity comparison only once both knockouts are confirmed to have removed what they targeted and to have changed the same number of atoms; the *size-matched coverage* is the share of the confirmed molecules that also meet this second condition. Locus, put to every alive feature, compares the drop from swapping one scaffold atom against the drop from swapping one non-scaffold atom; no group has to be found first, and nothing beyond the swap itself needs confirming, so this test reaches features the necessity test cannot.

For each feature, a two-sided Wilcoxon signed-rank test [30] is run across its gathered molecules, and the matched-pairs rank-biserial correlation r reports how one-sided the result is, from -1 , when the second edit’s drop always wins, to $+1$, when the first always does; each drop is expressed as a share of the feature’s own peak activation, the strongest it ever reaches on any molecule that switches it on, so the comparison never mixes features firing at different strengths. For necessity, the first edit is the target-group knockout: a significant positive r means the group cost the feature more than the control did, and the group is judged necessary. For locus, the first edit is the scaffold-atom swap: a significant positive r marks a feature read mainly from the scaffold core, a significant negative r one read from the atoms around it.

Neither test can be put to a feature with too few molecules to compare. A feature yielding fewer than six usable pairs is left untested rather than forced through regardless, six being the fewest paired differences at which the exact two-sided Wilcoxon test can ever reach significance at the conventional level. Within each test, the resulting p -values are corrected across all of a model’s features by the Benjamini–Hochberg procedure [35], and a result counts as significant once its corrected q -value falls below 0.05.

3. Results

3.1. What the Features Are, and How They Divide

The features we examine come from a sparse autoencoder trained separately on each of the six models, with its two settings, the number of features it forms and how many switch on for any one molecule, chosen per model. This gives, for every model, a dictionary of features to read. Not all of them carry anything to read: a feature that never switches on has no activating molecules to describe it, and nothing to contribute. We set these silent features aside as dead and keep the rest, the alive features, those that switch on for at least one molecule, as the object of everything that follows, reported on the `test` set. Dead features

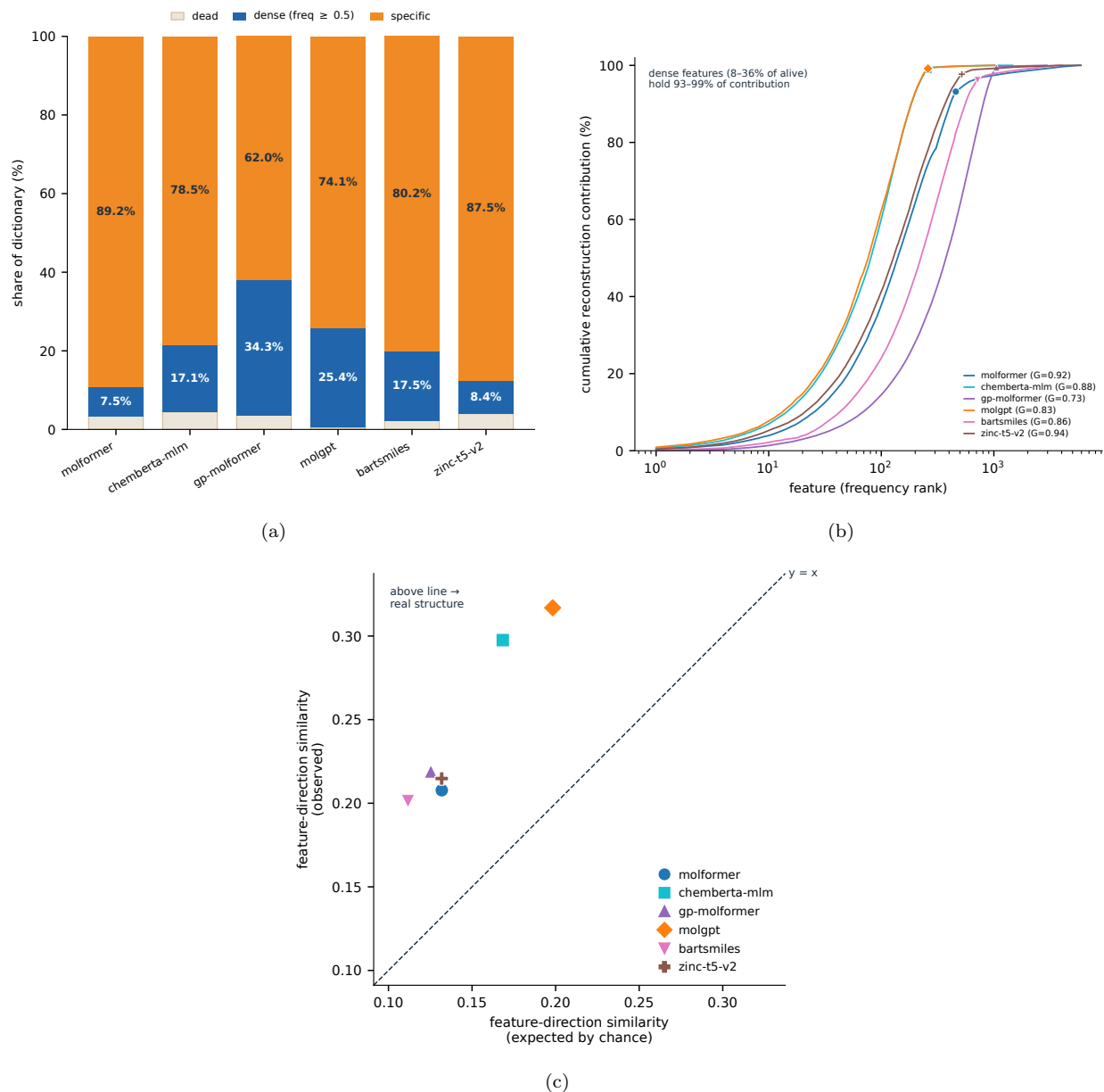


Figure 4: **The feature dictionaries split into two occupancy regimes.** (a) Share of dictionary features that are dead, dense (firing frequency ≥ 0.5), and specific, per model. (b) Cumulative reconstruction contribution against frequency rank (marker, dense-block boundary; legend, Gini coefficient). (c) Observed versus chance similarity of neighbouring feature (decoder) directions, per model.

Table 3: **Composition, rebuilding work and geometry of the six feature dictionaries.** Measured on the MOSES *test* set.

Model	Features in the dictionary					Rebuilding work		Closeness to nearest feature	
	Total	Dead	Alive	Dense	Specific	Features carrying half	Held by dense (%)	Observed	By chance
MoLFormer	6,144	207	5,937	459	5,478	139	93.2	0.21	0.13
ChemBERTa-MLM	1,536	68	1,468	262	1,206	80	98.8	0.30	0.17
GP-MoLFormer	3,072	112	2,960	1,054	1,906	368	99.5	0.22	0.13
MolGPT	1,024	5	1,019	260	759	74	99.2	0.32	0.20
BARTSmiles	4,096	96	4,000	716	3,284	216	96.2	0.20	0.11
ZINC-t5-v2	6,144	247	5,897	519	5,378	126	97.7	0.21	0.13

were few, between 0.5% and 4.4% across the six models, so almost every feature was alive, leaving roughly 1,000 to 6,000 usable features per model (Table 3).

These alive features are not all alike, and the first thing that separates them is how often each switches on. For every feature we measured this directly: across all the molecules in the set, the fraction that switch it on, its firing frequency. Most features were selective, switching on for only a small share of molecules, but a minority switched on for a great many. We divide the alive features at the half-way mark: a feature that switches on for at least half of all molecules we call dense, and the rest, the selective majority, specific (Fig. 4a; the full distribution of firing frequencies is shown for each model in Supplementary Fig. 6). The half-way cut is a reporting boundary rather than a sharp divide in the data, and we examine its exact value in Methods. Dense features were the minority in every model, making up between 7.7% (MoLFormer) and 35.6% (GP-MoLFormer) of alive features, with the selective specific features accounting for the remaining 64.4% to 92.3%. The two groups behave differently enough, as the next paragraphs show, that we carry this dense-or-specific label through every later analysis, as an annotation on each feature, never as a filter, since every alive feature is analysed regardless of which group it falls in.

If the two groups differ in number, they differ far more in weight. Beyond how often it fires, each feature does part of the work of rebuilding a molecule’s representation, and how much it contributes depends on how strongly it switches on together with the length of the direction it adds back. We ranked the alive features by this contribution and found it very unevenly shared: half of the total came from as few as 74 features in one model and no more than 368 in any, out of the roughly one to six thousand alive in each (Fig. 4b). Those few heavy features were almost exactly the broadly-firing ones, ordering features by contribution and by firing frequency gave nearly the same ranking (Spearman correlation of 0.96 or higher), so it was the dense minority, already set apart by its firing, that carried the load. As a share, the dense features held between 93.2% and 99.5% of the total contribution in every model, leaving the far larger specific majority with only the small remainder, 0.5% to 6.8%. Weight and number therefore run in opposite directions: a handful of dense features do most of the reconstruction, while each of the many specific features adds only a little.

Throughout, we have spoken of each feature as a separate part standing for its own pattern, which only holds if the dictionary really is a collection of distinct directions rather

than many near-copies of a few. We checked this by looking at the directions the features add back when rebuilding a representation. Because each model forms several times more features than its representation has dimensions, these directions cannot all point independently, and some overlap is unavoidable; we therefore compared every measurement against the same number of purely random directions as a baseline. Each feature’s direction lay somewhat closer to its nearest neighbour than random directions would, a closeness of 0.20 to 0.32 across models, against 0.11 to 0.20 for the random baseline, on a scale where 1 means identical and 0 unrelated, yet almost none were near-copies: in every model fewer than half a percent of features had a neighbour closer than 0.8 (Fig. 4c). Taken together, the features still spread across a large share of the available directions, roughly half to two-thirds of the representation’s dimensions, a little less than fully random directions would but nowhere near collapsing onto a few. The dictionary is therefore a set of genuinely distinct directions, which is what lets us read each feature on its own in the analyses that follow.

We now hold, for each model, a dictionary of genuinely distinct features, sorted into a small dense minority that does most of the work of rebuilding a molecule and a large specific majority that each does little, with that label kept on every feature as we go. What none of this yet tells us is what any single feature actually responds to, which piece of chemistry switches it on. That is the question we turn to next, by gathering the molecules that most strongly activate each feature and asking what structure they share.

3.2. What Each Feature Is Associated With

Table 4: **Structural associations of each feature, by granularity.** Features are those alive on the MOSES *scaffold-test* set molecules; the standout is a feature’s most over-represented structure, and carriers are the molecules that hold it.

Model	Features	With an association (%)		Structures per feature		Odds for the standout		Carriers switching on (%)	
		Specific	Dense	Specific	Dense	Specific	Dense	Specific	Dense
<i>Functional groups</i>									
MoLFormer	5,896	98.1	100.0	17	39	30	46	17.0	100.0
ChemBERTa-MLM	1,403	90.3	100.0	11	29.5	58	37	6.7	100.0
GP-MoLFormer	2,958	99.8	99.6	16	25	48	24	15.1	100.0
MolGPT	1,018	99.9	94.6	17	29	29	29	13.5	100.0
BARTSmiles	3,991	98.9	97.2	23	25	30	30	27.3	100.0
ZINC-t5-v2	5,632	85.0	99.2	12	53	50	35	8.8	100.0
<i>Scaffolds</i>									
MoLFormer	5,896	97.5	71.5	24	253.5	189	125	60.0	100.0
ChemBERTa-MLM	1,403	82.5	64.5	12	69	206	68	35.7	100.0
GP-MoLFormer	2,958	99.0	71.4	15	12	166	36	44.4	100.0
MolGPT	1,018	99.3	55.0	23	48	107	53	40.9	100.0
BARTSmiles	3,991	98.7	54.8	37	30	131	54	64.9	100.0
ZINC-t5-v2	5,632	80.6	75.7	10	178	212	75	35.0	100.0

To find what a feature responds to, we take the molecules that switch it on and ask what they share that the rest do not, reported on the **scaffold-test** set. These are the held-out molecules, whose scaffolds the autoencoders never saw during training. We test

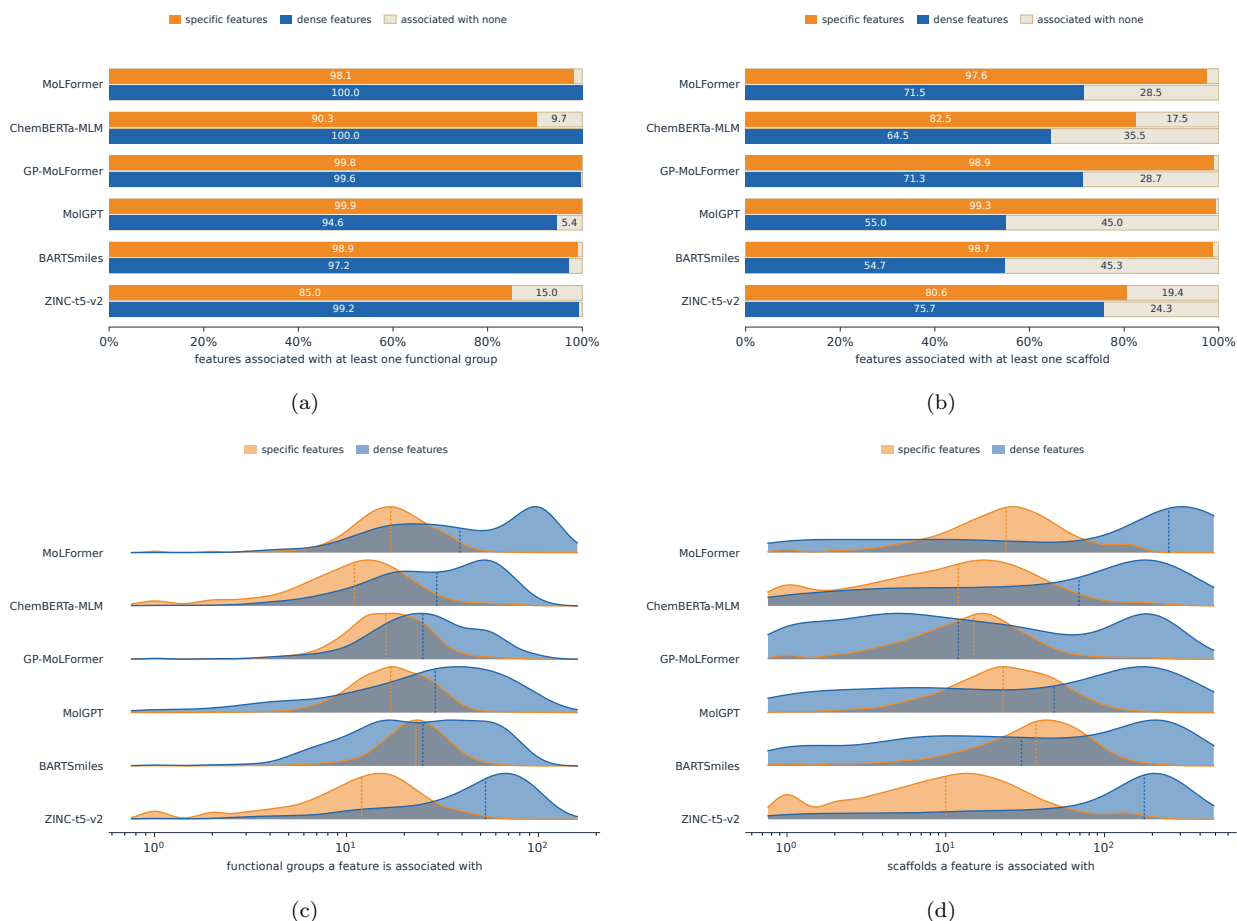


Figure 5: **Structural associations of each feature.** Functional groups, left; scaffolds, right. Each feature was tested against all 650 in-vocabulary scaffolds and all 180 functional-group types, comparing the molecules that switch it on against the rest of the corpus (one-sided Fisher exact test, per-feature Benjamini–Hochberg [35] correction, $q < 0.05$ at a non-negative $\log_2$ odds ratio). **(a, b)** Features associated with at least one structure, as a share of each regime. **(c, d)** Structures per feature, among the features carrying at least one; curves, distribution over features; dashed rule, median.

each feature against every structure common enough to compare against, those appearing in at least thirty of these molecules: 180 functional-group types and 650 scaffolds. A structure counts as associated with a feature when it appears among that feature’s molecules more often than chance allows, corrected for the many structures each feature is tested against. A few features alive on the molecules of the previous section fall silent on these, at most 4.5% in any model. Nearly every one that remained was associated with a functional group, 86.3% (ZINC-t5-v2) to 99.7% (GP-MoLFormer) across the six models (Table 4), and this held for both groups: 85.0% to 99.9% of specific features and 94.6% to 100% of dense ones (Fig. 5a). Scaffolds are harder to reach here, since holding them out leaves most molecules with a near-unique one and only 44.8% carry a scaffold common enough to compare against. Specific features found one anyway, 80.6% to 99.3%. Dense features less often, 54.7% (BARTSmiles) to 75.7% (ZINC-t5-v2) (Fig. 5b).

Features that stood for one piece of chemistry would each be associated with one structure and no other. Almost none were. The typical specific feature was associated with 11 (ChemBERTa-MLM) to 23 (BARTSmiles) functional groups, and the typical dense feature with 25 to 53 (Fig. 5c). Scaffolds gave the same picture for specific features, 10 (ZINC-t5-v2) to 37 (BARTSmiles) (Fig. 5d). Dense features varied most here, from a typical 12 in GP-MoLFormer to 253.5 in MoLFormer, where a tenth were associated with a single scaffold and three-quarters with more than ten. Whatever a feature responds to, it is not one piece of chemistry.

Among the structures a feature was associated with, one stood out above the rest, the one most over-represented among the molecules that switch it on. These were not a few common structures recycled across features: 173 to 180 of the 180 functional-group types, and 375 to 633 of the 650 scaffolds across models, were the standout structure of at least one feature. Each of these associations was strong. For a specific feature, the odds of switching on, the chance it fires set against the chance it does not, were 29 (MolGPT) to 58 (ChemBERTa-MLM) times higher among molecules carrying its standout functional group than among molecules without it, and 24 to 46 times higher for dense features (Fig. 6a). For scaffolds the same comparison ran higher in every model, 107 (MolGPT) to 212 (ZINC-t5-v2) times for specific features and 36 to 125 for dense ones (Fig. 6b). Yet the standout was the largest of many rather than a structure apart: across features of both kinds it held 6% to 19% of a feature’s total association at functional-group level and 1% to 15% at scaffold level.

Each measurement so far has counted structures against features. A different question is what carrying one of those structures does to a molecule: across the whole set, how many molecules switch a feature on, and among only those carrying its standout structure, how many? For specific features the two are far apart. These features switch on for 0.3% to 1.6% of molecules across the set, but for 6.7% (ChemBERTa-MLM) to 27% (BARTSmiles) of molecules carrying their standout functional group (Fig. 6c) and 35% (ZINC-t5-v2) to 65% (BARTSmiles) of those carrying their standout scaffold (Fig. 6d). Carrying the structure changes whether these features switch on. Dense features show the reverse. They already switch on for 86% to 99% of molecules across the set, and the typical one switches on for every molecule carrying its standout structure, so the structure changes almost nothing, because almost nothing is left to change. This is why so many dense features had no standout scaffold at all: a feature that switches on for nearly everything cannot single anything out.

Across all six models, then, almost every feature is tied to chemical structure, each to

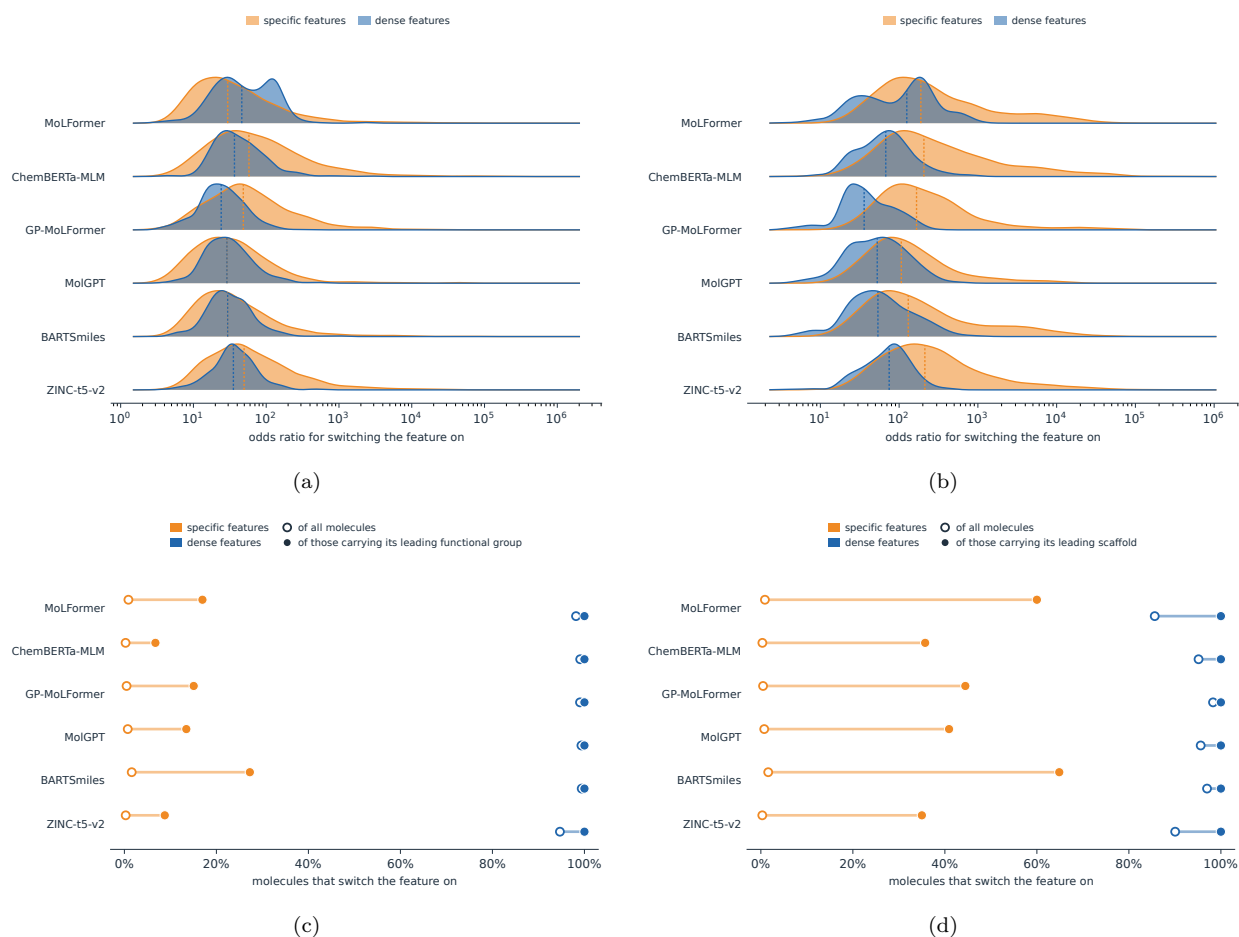


Figure 6: **Magnitude of each feature’s leading association.** Each feature contributes its single most over-represented structure. Functional groups, left; scaffolds, right. **(a, b)** Odds of switching the feature on for a molecule carrying that structure, relative to molecules without it; curves, distribution over features; dashed rule, median. **(c, d)** Molecules that switch a feature on, as a share of all molecules (open marker) and of those carrying its leading structure (filled marker); medians across features.

many structures at once, one of them the largest, and for specific features the presence of that structure changes whether the feature switches on. Every one of these is a count of molecules in which a feature and a structure appear together. A structure that appears alongside a feature need not be one the feature depends on; it may simply travel with whatever the feature is actually reading, and nothing measured so far tells the two apart. Doing so takes a different kind of evidence: take the structure out of a molecule that switches the feature on, and see whether the feature falls quiet or fires just the same. Where in the molecule that change is made answers a second question, whether a feature draws on the central ring framework or on the atoms around it. We turn to both next.

3.3. *What Each Feature Actually Needs, and Where it Reads*

A structure cannot be cut out of a molecule; what remained would no longer be a molecule. Instead we change one atom into another. A functional group is defined by the elements in it, so turning one of them into a different element takes the group away and leaves everything else standing, the same atoms joined in the same way. A scaffold is the central ring framework itself, and removing it would leave no molecule behind, so the question is put to functional groups alone. Because any change unsettles a molecule a little, each removal is paired with a control, the same kind of change made elsewhere in the same molecule, which leaves the group in place. Sent back through the model, a feature that reads the group falls further with it gone than under the control; one that merely fired alongside it falls the same either way. A Wilcoxon signed-rank test [30] asks whether one drop is consistently the larger across the molecules that switch the feature on most strongly, corrected across all of a model’s features by Benjamini–Hochberg [35]; with fewer than six molecules to compare, nothing can be told apart from chance and the feature is left undecided. Not every group can be reached. Carbon, nitrogen, oxygen and sulfur each hold several bonds and can stand in for one another, while fluorine, chlorine and bromine hang off the molecule by a single bond, so exchanging one would change how the molecule is joined rather than only what it is made of. Between 10.7% (GP-MoLFormer) and 19.1% (ChemBERTa-MLM) of the features carrying a standout functional group therefore have no test to take. What falls outside is a particular chemistry, not a random sample, and every share below is a share of the features the test reaches.

Put to the test, a little over half the readings held. Among the specific features the test could decide, removing the standout group cost the feature more than the control did in 54.6% (BARTSmiles) to 65.2% (GP-MoLFormer) (Table 5), marking the group as one the feature reads (Fig. 7a). For dense features the share was lower, 43.6% (MoLFormer) to 57.4% (GP-MoLFormer) (Fig. 7b). The rest failed in two ways. Most fell the same either way, 26.7% to 33.9% of specific features and 19.5% to 36.6% of dense ones, so the group was present in every molecule that switched the feature on and the feature was not reading it. A smaller share fell further under the control than under the removal, 8.1% to 13.3% of specific features and 16.6% to 25.5% of dense ones, so what these features read lay elsewhere in the molecule and the removal left it untouched. A last share could not be tested at all, for want of six molecules that both switch the feature on and carry its group, and this fell almost entirely on the specific features, 19.2% (BARTSmiles) to 35.7% (ZINC-t5-v2) against 0.5% to 2.2% of dense ones, which follows from how rarely they fire: a feature switching on for one molecule in a hundred has far fewer to draw on than one switching on for nearly all. Taking

Table 5: **What each feature needs, and where it reads.** Necessity is put to functional groups only; percentages are of the preceding count column, and activation lost is a median share of a feature’s peak firing.

<i>Does the feature need its standout functional group?</i>								
Model	Type	Features		Of those tested (%)			Activation lost (%)	
		Removable group	Tested	Reads its group	Neither	Reads elsewhere	Its group	Control
MoLFormer	Specific	4,561	3,661	57.0	30.9	12.1	23.3	8.3
	Dense	413	404	43.6	36.6	19.8	6.5	4.3
ChemBERTa-MLM	Specific	823	565	60.9	28.3	10.8	19.7	6.4
	Dense	222	218	52.8	27.5	19.7	4.1	1.4
GP-MoLFormer	Specific	1,668	1,214	65.2	26.7	8.1	42.8	9.3
	Dense	965	958	57.4	25.5	17.1	11.5	7.1
MolGPT	Specific	657	495	63.0	27.3	9.7	35.7	7.2
	Dense	222	220	55.0	19.5	25.5	8.1	5.4
BARTSmiles	Specific	2,769	2,237	54.6	32.1	13.3	17.0	4.0
	Dense	595	592	50.5	28.0	21.5	4.1	2.0
ZINC-t5-v2	Specific	3,522	2,266	57.7	33.9	8.4	30.8	11.3
	Dense	452	447	51.9	31.5	16.6	7.3	3.2

<i>Where on the molecule does the feature read?</i>								
Model	Type	Features		Of those tested (%)			Activation lost (%)	
		Alive	Tested	Outer atoms	Neither	Central frame	Outer atoms	Central frame
MoLFormer	Specific	5,429	5,322	57.9	22.8	17.3	23.7	12.5
	Dense	459	459	76.5	15.9	7.6	15.8	7.2
ChemBERTa-MLM	Specific	1,118	995	52.4	26.2	10.4	25.7	11.0
	Dense	262	262	76.3	17.6	6.1	12.9	4.8
GP-MoLFormer	Specific	1,888	1,875	65.1	24.1	10.1	35.5	19.2
	Dense	1,052	1,050	88.8	10.2	0.9	29.3	15.4
MolGPT	Specific	756	754	79.5	14.4	5.8	39.9	17.2
	Dense	260	260	93.8	5.0	1.2	23.6	6.8
BARTSmiles	Specific	3,270	3,233	58.6	26.2	14.1	16.4	5.4
	Dense	716	716	70.7	20.7	8.7	11.9	3.4
ZINC-t5-v2	Specific	5,067	4,290	44.6	30.0	10.1	29.3	19.3
	Dense	519	519	75.3	20.2	4.4	21.5	12.0

both groups together, 38.2% (GP-MoLFormer) to 46.2% (BARTSmiles) of the associations the test could decide did not survive the removal of the structure they rested on. Counting a structure alongside a feature, however strongly, does not establish that the feature needs it.

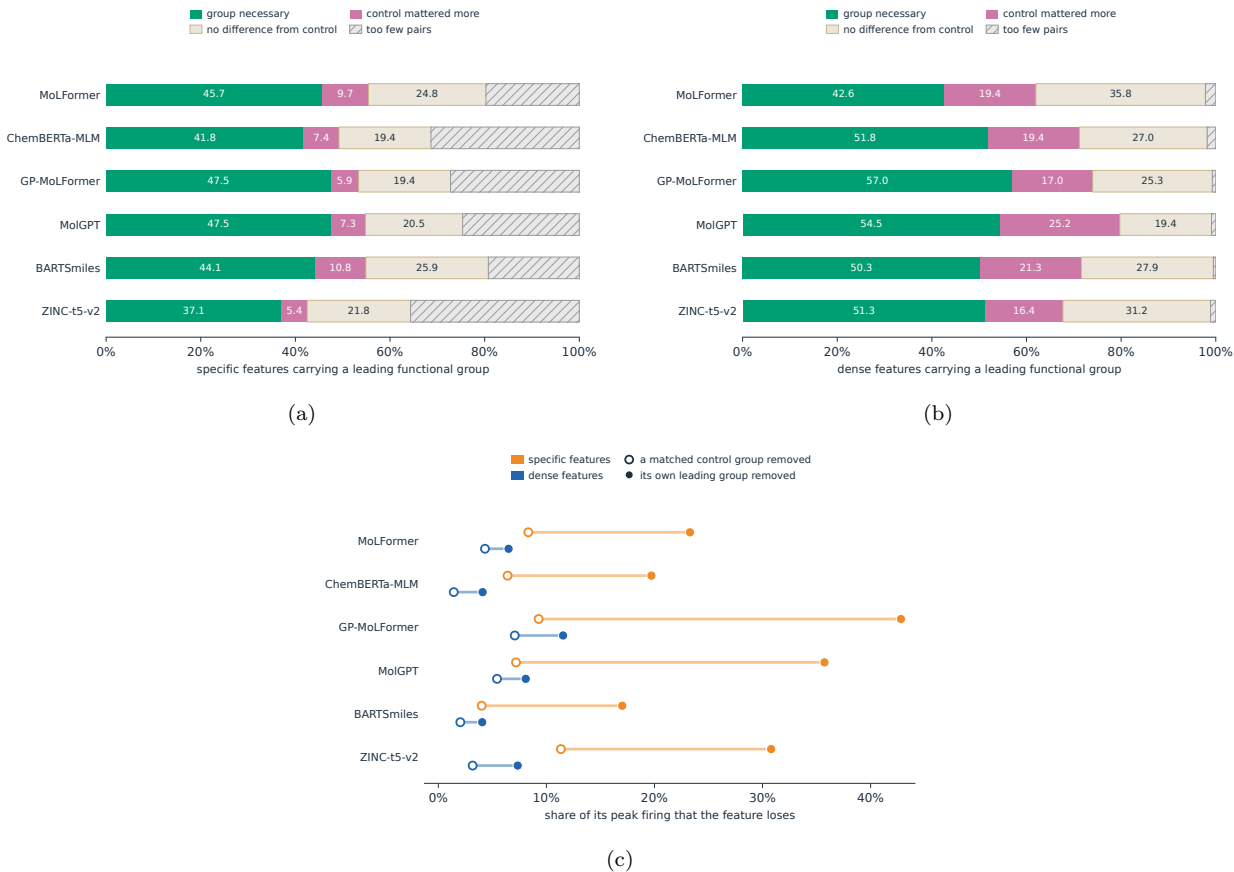


Figure 7: **Necessity of each feature’s leading functional group.** The group is removed against a matched control in the same molecule; paired drops are compared by two-sided Wilcoxon signed-rank test, per-feature Benjamini–Hochberg, $q < 0.05$. Features with fewer than six usable pairs are untested. **(a, b)** Features carrying a leading functional group, by outcome; specific, **(a)**, and dense, **(b)**, each bar normalised to its own regime. **(c)** Median activation lost under each edit, as a share of the feature’s peak firing, across features the test decided.

A verdict says which of the two changes cost the feature more, not how much it lost. Features fire at different strengths, so each drop is set against the feature’s own peak, the strongest it ever fires on any molecule that switches it on. For a specific feature, removing the standout group cost 17.0% (BARTSmiles) to 42.8% (GP-MoLFormer) of that peak, against 4.0% to 11.3% for the control (Fig. 7c). Taking the group away removes a substantial part of what these features had to give. Dense features fell the same way round in every model, the group costing more than the control, but both losses were small: 4.1% to 11.5% for the group against 1.4% to 7.1% for the control. This is the ceiling met in the previous section from the other side. A dense feature already switches on for nearly every molecule and sits near the top of its range, so there is little room left to fall, whatever is taken away. And these are the features carrying nearly all the work of rebuilding a molecule, 93% or more

of it in every model. The parts of the model doing the most work are the parts chemistry moves the least. A dense feature can pass the test and still lose only a few percent of its range when the group it reads is removed, so surviving the test and mattering to the feature are not the same thing.

One of those two failures needs a second look. A feature that fell the same either way may not be reading the group, or the test may simply have had too few molecules to tell. Sorting the tested features by how many molecules the test had separates the two. For specific features the share found to need their group climbs steadily with the evidence available, from 21% (MolFormer) to 36% (MolGPT) where only six to ten molecules could be compared, to 80% (ChemBERTa-MLM) to 91% (GP-MolFormer) where more than a hundred could (Supplementary Fig. 7). Much of that band is therefore a shortage of molecules rather than an absence of dependence, and the shares reported above are a floor. Dense features do not behave this way. Given the same hundred molecules and more, they are found to need their group in only 52% (BARTSmiles) to 57% (GP-MolFormer) of cases, no better than at a fifth of the evidence. Where a specific feature converts once the molecules are there, a dense feature does not, however many it is given. The same outcome thus means two different things: for a specific feature, usually that the test could not see far enough; for a dense one, that there was nothing to see.

Necessity asks whether a group matters. It says nothing about where on a molecule a feature is reading. The same kind of change answers that, moved rather than removed: one atom is changed on the scaffold, the central ring framework a molecule is built around, and one on the atoms hanging off it, in the same molecule. Nothing has to be taken away for this, and no standout group is needed, so every alive feature can be asked, including those with no group the removal test could reach. The answer came out the same in every model. Among specific features, the atoms around the frame won in 44.6% (ZINC-t5-v2) to 79.5% (MolGPT) and the core in 5.8% (MolGPT) to 17.3% (MolFormer), the rest favouring neither or having too few molecules to compare (Fig. 8a). Dense features leaned further out still, 70.7% to 93.8% against 0.9% to 8.7% (Fig. 8b). Set against each other, of the features that favoured either side the surrounding atoms took 77.1% to 93.2% of specific ones and 89.1% to 99.0% of dense ones, and 78.3% to 94.7% with both groups pooled. The amounts lost agree with the verdicts. Changing an atom outside the core cost a specific feature 16.4% (BARTSmiles) to 39.9% (MolGPT) of its peak, against 5.4% to 19.3% for a change on the core, and a dense feature 11.9% to 29.3% against 3.4% to 15.4% (Fig. 8c). The outer change cost more in every model and in both groups. The scaffold is how a chemist names a molecule’s family, and it is what the held-out split was built on. It is not where these features read. What moves them is what hangs off the frame.

The same edit has now answered two questions, and the answers agree. Between 38.2% and 46.2% of the associations the test could decide did not survive the removal of the structure they rested on, and of the features that favoured either side of a molecule, 78.3% to 94.7% read from the atoms around the scaffold rather than from the scaffold itself. The two groups of features carried since the first subsection part company at every step. A specific feature loses several times more of its peak when its own group goes than when the control does, is found to need its group once the test has enough molecules to see the difference, and reads from the outer atoms in most cases. A dense feature moves the same way at every step but by a fraction of the amount, does not come round however many molecules it is

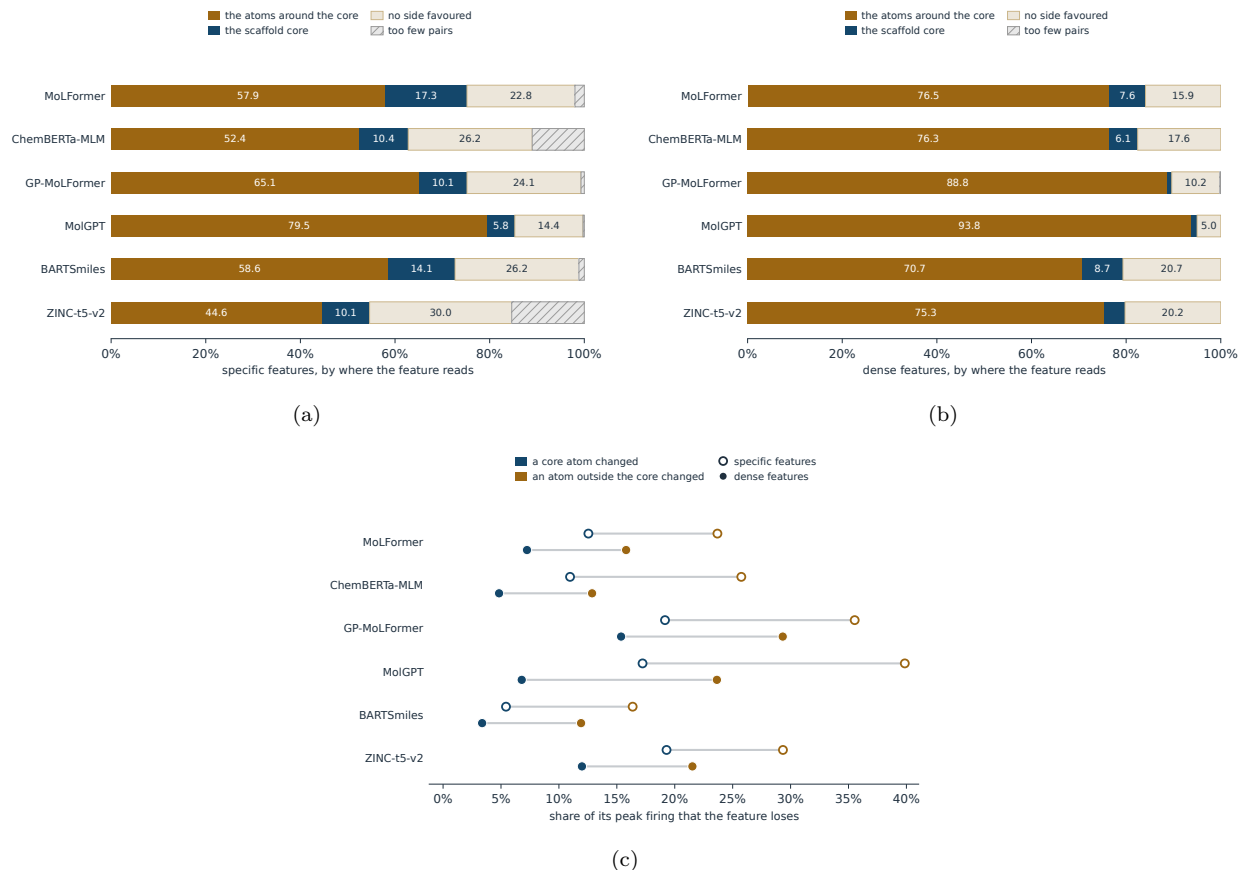


Figure 8: **Locus of each feature’s activation.** One atom is changed on the Bemis–Murcko scaffold core against one outside it, in the same molecule; paired drops are compared as in Fig. 7, and the sign of the effect sets the side. Every alive feature is tested. **(a, b)** Alive features by the side favoured; specific, **(a)**, and dense, **(b)**, each bar normalised to its own regime. **(c)** Median activation lost under each swap, as a share of the feature’s peak firing, across features the test decided.

given, and reads from the outer atoms almost to the exclusion of the core. Yet it is these dense features, few as they are, that carry 93% or more of the work of rebuilding a molecule. What does most of the model’s work responds least to what a molecule is made of. Every association in the previous subsection was a count of molecules in which a feature and a structure appeared together. Putting each one to an edit turns it from something observed into something tested, and a substantial share does not survive. Reading a feature from what its strongest molecules share is where interpretation begins; it is not where it can be left.

4. Conclusion

Mechanistic interpretability in computational chemistry has historically relied on identifying the shared structural patterns that activate a model’s internal features. However, this study demonstrates that observational association is an insufficient metric for understanding chemical language models; when subjected to direct causal testing via Interpret MolT, between 38% and 46% of statistically significant feature-to-structure correlations fail. A feature’s tendency to fire alongside a functional group does not guarantee that the feature actually requires that group to function.

Furthermore, this causal interrogation exposes a critical dichotomy in how molecular transformers distribute their computational workload. These models are functionally anchored by a sparse minority of “dense” features, which perform over 93% of the representational reconstruction but exhibit the weakest causal reliance on the specific chemical structures they correlate with. Conversely, the “specific” features—which genuinely depend on their associated structures and primarily read from peripheral atoms rather than the central scaffold—contribute only a fraction of the reconstruction weight. Ultimately, a model’s most functionally consequential features are often the ones whose structural readings can be trusted the least.

For artificial intelligence to be reliably deployed in the high-stakes environment of drug discovery, interpretability claims must move beyond surface-level correlation. Pairing observational structural associations with explicit, counterfactual necessity tests provides a necessary and rigorous foundation for auditing molecular transformers. Only by proving that a feature’s activation is causally dependent on a specific structural sequence can we confidently claim to know what a chemical language model has learned.

Data availability

The molecular data analysed in this study is from the publicly available MOSES benchmark [24], distributed with its standard train, test and scaffold-test splits. The six pretrained molecular transformers are publicly available from their original repositories [5, 6, 25, 26, 27, 28].

Code availability

Code for embedding extraction, sparse-autoencoder training and selection, structural association, and the counterfactual causal-testing analyses is available from the corresponding author on request.

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CRedit authorship contribution statement

- **D.U.K:** Conceptualization, Methodology, Software, Writing – original draft, Visualization
- **N.A:** Methodology, Validation, Investigation, Writing – original draft
- **F.H:** Resources, Visualization
- **A.D:** Supervision, Writing – review and editing
- **M.N.A:** Conceptualization, Writing – review and editing, Supervision

Declaration of competing interests

The authors declare no competing interests.

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